

Cognitive Modeling of Creativity in Drama-Based Teacher Education: Insights from AI-Assisted Reflection

Yaxin Hu

*Applied Language Studies
Education and Languages School
Hong Kong Metropolitan University;
Hong Kong, China
Foreign Language School
Wuhan City Polytechnic,
Wuhan, China*

Jack Shu

s1350034@live.hkmu.edu.hk Jack Shu
*Applied Language Studies
Education and Languages School
Hong Kong Metropolitan University
Hong Kong, China
jshu@hkmu.edu.hk*

Abstract—This research focuses on how the AI-powered reflection can contribute to the cognitive modeling of creativity in the teacher education programs involving drama. Grounded in creativity theories by Guilford, Sawyer, and Robinson, it explores how pre-service teachers engage in imaginative, transformative, and expressive cognitive acts during children's theatre projects. Using reflective writing, dialogue transcripts, and AI tools such as generative language models and video annotation systems, this study builds a conceptual framework for tracking and analyzing the stages of creative cognition. The findings demonstrate that AI-assisted reflection fosters meta-cognitive awareness, encourages iterative thinking, and enhances personalized creative development. The study proposes a cognitive pathway model of creativity within drama pedagogy and discusses implications for future AI-integrated teacher training programs.

Keywords—Creativity, Drama Pedagogy, Cognitive Modeling, AI-Assisted Reflection, Teacher Education, Generative AI

I. INTRODUCTION

In recent years, the integration of artificial intelligence (AI) in education has shifted from content delivery to personalized learning and reflective thinking [1], [2]. At the same time, drama-based pedagogy has gained prominence in teacher education for its potential to cultivate creativity, empathy, and embodied understanding. However, the cognitive processes underpinning creative acts in drama education remain underexplored, particularly in the context of AI-assisted learning environments. This study aims to address this gap by investigating how AI tools can support the cognitive modeling of creativity among pre-service teachers engaged in drama-based learning.

II. THEORETICAL FRAMEWORK

Creativity has long been conceptualized as a multidimensional construct. Guilford [3], [4] defined it primarily in terms of divergent thinking, emphasizing fluency, flexibility, originality, and elaboration as measurable cognitive abilities. While influential, this cognitive model has been criticized for treating creativity as an individual trait detached from social and contextual factors. In the context of

drama pedagogy, Guilford's focus on divergent idea generation is useful but insufficient, as it does not explain how collaborative improvisation and embodied performance contribute to creative outcomes.

Sawyer [6], [7] extended sociocultural theory to highlight collaborative emergence. From this perspective, creativity is not only generated within individuals but also emerges dynamically from group interaction. Drama activities are all about improvising, teaming up, and co-constructing the stories. These kinds of drama activities really picture Sawyer's understanding of creativity among the distributed and emergent. This mindset, contrary to Guilford's emphasis on cognition, which can only offer a partial explanation of how students in drama develop collaboratively, is what this model is all about.

According to Robinson and Aronica [5], creativity is not just an individual ability but an attribute every thoughtful person can develop in their everyday life. Their work emphasizes that learning spaces need to be designed in such a way that requires that sort of taking risks and different expressions, which are not focused only on outcomes defined by standards. Such a pre-service teaching program is much in line with teacher education, where pre-service teachers should not only be creative individually but also foster creativity in the classes of the schools they teach in one day.

Cognitive modeling is the approach that facilitates the connection between the two views. Following the discussion of Crapo et al. [2], cognitive simulation includes the idea of replicating mental activities that are usually illustrated through sequences, steps, or pathways. In this study, we conceptualize creativity as both cognitive and socially situated. We argue that Guilford's divergent thinking is best understood as the starting point of imaginative projection, Sawyer's collaborative emergence explains transformative synthesis in group rehearsals, and Robinson's emphasis on creative confidence frames expressive externalization. This theoretical triangulation provides a foundation for the proposed cognitive pathway model and clarifies how AI-assisted reflection can support different stages of creative development.

III. METHODOLOGY

A. Participants and Context

The study was conducted with 18 pre-service teachers enrolled in a drama-based educational methods course at a Chinese vocational university. Students worked in small groups to devise and perform children's theatre projects over a 12-week semester. While this small and context-specific sample enabled in-depth qualitative analysis, we acknowledge that it constrains the generalizability of the findings.

B. Data Collection

To capture the cognitive processes underlying creativity, we employed three qualitative data sources across the 12-week learning cycle:

1) Reflective Journals

Each participant submitted weekly reflective journals documenting evolving ideas, challenges, decision-making strategies, and emotional responses. These journals were composed with the assistance of GPT-4 (off-the-shelf, not fine-tuned), which provided iterative prompts, rephrasings, and reflective questions. This AI-supported interaction encouraged participants to reconsider narrative structures and explore alternative creative possibilities, producing a rich corpus of metacognitive data.

2) Dialogue Transcripts

Group rehearsals and planning sessions were recorded and transcribed using Otter.ai. The transcripts were then analyzed with NLP-based tagging and clustering:

- To identify salient concepts, keyword extraction combined TF-IDF with neural embeddings. TF-IDF highlighted terms with high discriminative power, while neural embeddings grouped semantically related words (e.g., 'improvise' and 'improvisation') into coherent categories.
- To detect conceptual patterns, we applied semantic clustering with sentence-transformer embeddings (all-MiniLM-L6-v2). This model converts each sentence into a numerical vector, allowing reflections with similar meanings (e.g., expressions of doubt, reformulation, or resolution) to be automatically grouped together.
- To examine how creativity unfolded over time, we conducted temporal mapping of key events across the rehearsal timeline. Python libraries such as Pandas and Scikit-learn were used to align timestamps and detect patterns (e.g., whether divergent idea generation peaked in early rehearsals, or expressive embodiment emerged closer to final performances).

To enhance validity, a subset of AI outputs was manually reviewed by the researchers. Discrepancies were discussed and resolved, ensuring that AI-assisted coding complemented rather than replaced human interpretation.

3) Performance Video Annotations

We Final performances and selected rehearsals were analyzed using ELAN 6.4 for multimodal annotation. Annotations focused on body movement, vocal shifts, spatial use, and facial expressions that embodied creative processes. These visual data provided a

complementary perspective and supported multimodal triangulation.

C. Data Analysis

A constructivist grounded theory approach [1] guided the analysis. Human coders first conducted open coding to identify units of meaning (e.g., role reshuffling, idea deviation, emotion shifts), which were grouped into broader categories corresponding to stages of creative cognition: imaginative projection, transformative synthesis, and expressive externalization.

To support and extend human coding, several AI-assisted techniques were employed:

1) Keyword Extraction

TF-IDF weighting was combined with neural embeddings to highlight salient terms. TF-IDF identified words with high discriminative power, while embeddings grouped semantically related words (e.g., improvise and improvisation) into coherent categories.

2) Semantic Clustering

Sentence-transformer embeddings (all-MiniLM-L6-v2) were used to represent each sentence as a vector, enabling reflections with similar meanings to be automatically grouped together. This helped reveal conceptual trails such as cycles of doubt, reformulation, and expression.

3) Temporal Mapping

Python libraries (Pandas, Scikit-learn) were used to align timestamps and detect when specific creative events occurred in the rehearsal process. This made it possible to observe, for example, whether divergent idea generation peaked during early rehearsals or whether expressive embodiment became more prominent near final performances.

4) Triangulation and Human Oversight

Patterns identified by AI tools were cross-checked with human coding across journals, transcripts, and video annotations. In cases of discrepancy, results were discussed and refined collaboratively. This hybrid workflow ensured that AI-assisted coding complemented, rather than replaced, human interpretation, thereby strengthening the trustworthiness of the findings.

IV. FINDINGS

A. Stages of Creative Cognition in Drama Pedagogy

Analysis revealed three recursive stages of creative cognition:

1) *Imaginative Projection*: idea generation and world-building, often characterized by divergent thinking and playful experimentation.

2) *Transformative Synthesis*: integrating texts, cultural references, and group perspectives to converge ideas into a coherent dramatic structure.

3) *Expressive Externalization*: embodying and communicating ideas through performance, gesture, and voice.

These stages rarely followed a linear order. Instead, students frequently cycled between them, reflecting the iterative and embodied nature of creativity in drama pedagogy.

B. Role of AI in Fostering Creative Reflection

AI-supported reflection tools played a significant role in amplifying metacognitive awareness. Journals written with GPT-4 prompts encouraged students to revisit narrative structures from multiple perspectives. NLP-based analysis of dialogues revealed how collaborative meaning-making unfolded, while video annotations highlighted embodied expressions of creativity. Students reported that AI feedback—particularly prompts to “consider alternative viewpoints” or “clarify emotional intent”—helped them reframe challenges and generate new possibilities.

C. Cognitive Pathway Model of Creativity

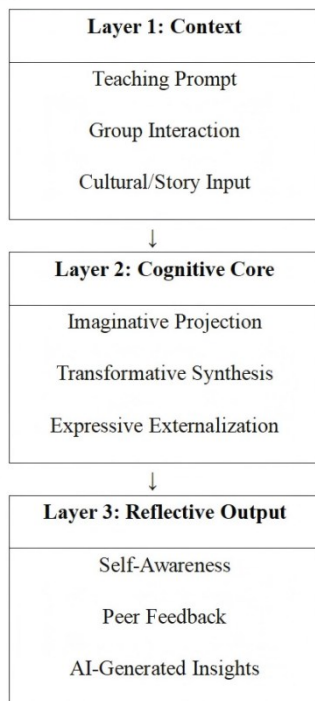


Fig.1. Cognitive pathway model of creativity

- From these findings, we propose a three-layered cognitive pathway model:
- Layer 1: Contextual Inputs (pedagogical prompts, group interactions, story resources)
- Layer 2: Cognitive Processes (imagination → transformation → expression)
- Layer 3: Reflective Outputs (self-awareness, peer commentary, AI-supported reflection)
- This model illustrates how creativity in drama education emerges from the dynamic interplay of context, cognition, and reflection, and how AI can serve as a mediating tool in this recursive process.

Figure 1 has three processes, which are conceptually linked to the projects being conducted by pre-service teachers in the drama-based area. The top-level layer (Layer 1) is a context layer, which is elaborated by learning tools, such as

prompts, group discussions, and narrative sources. These materials as inputs will be the storage devices where the inner cognitive processes (Layer 2), such as imagination, transformation, and the outside ability, will be invoked. The ultima layer (Layer 3) is that of reflecting outcomes, such as self-awareness, peer review, and AI's feedback, which are recursive processes that lead to improvement in metacognition.

This model helps show that the dramatic education creativity indeed has a physical substratum and a social fabric, but the top level is the metacognitive loops facilitated by AI.

V. DISCUSSION

A. Rethinking Creativity in Teacher Education

The results showed that creativity in drama pedagogy-based learning is not just an individual trait, but a more complex and iterative process, which is more dynamic and unpredictable. Guilford also proposed a model illustrating where the cognitive processes related to the divergent thinking occur [3]; however, he did not consider why and how it is possible to compose freely during improvisation. Collaborative emergence [6], shared by Sawyer, provides a stronger reason for individual improvisations and group negotiations working together, whereas Robinson's understanding of creative confidence [5] highlights the idea of success through risk-taking and positive reactions. Therein lies the rationale of drama education as creatively cognitive and socially contextualized typologies.

B. Consultation of AI as the Intelligent Dialogue Partner

The AI service, which engaged in this endeavor, was not intended to substitute for the human educators but for the reflective colleague. Unlike common approaches, where AI is represented in the education system as a means for automation to do grading essays, providing a feedback that is drill-based, or generating adaptive learning paths, our ones is showing the nontraditional potential of AI. There, AI as a tool to boost metacognition helped to make students question the narrative choices, knowing the emotional cues, and presenting the alternate interpretations. Such a transformation of AI from being a digital evaluator to a reflection partner of human mind not only as a replacement for a human judge but as an accelerator of deep reflection and creative risk-taking, will be possible.

C. Practical Implications

Three practical implications emerge for teacher education. First, embedding structured AI reflection checkpoints within drama courses can help pre-service teachers recognize their creative trajectories and cultivate reflective habits. Second, integrating multimodal reflection—across journals, dialogues, and performances—ensures that creativity is understood holistically rather than reduced to written text. Third, educators should promote responsible AI engagement, training students to critically evaluate machine-generated suggestions and maintain agency in decision-making.

D. Ethical Considerations and Limitations

This study involved 18 participants in a single institutional context, limiting the generalizability of findings. While the small sample enabled in-depth qualitative exploration, broader and cross-cultural studies are needed to

validate the model. Ethical considerations also arise in AI-assisted pedagogy, including issues of bias, transparency, and potential overreliance on automated suggestions. Human oversight remains essential, and educators must equip students with critical digital literacy skills to navigate AI responsibly.

E. Directions for Future Research

Future studies should explore multimodal AI systems capable of recognizing gestures, tone, and affective states to capture embodied creativity more comprehensively. Longitudinal designs could further reveal how AI-assisted reflection influences teacher identity, classroom practice, and sustained creativity development.

VI. CONCLUSION

This study examined how AI-assisted reflection can support the cognitive modeling of creativity in drama-based teacher education. By analyzing reflective journals, group dialogues, and video performances, we identified three recursive stages of creative cognition—imaginative projection, transformative synthesis, and expressive externalization. These stages, while distinct in analysis, often overlapped and cycled in practice, demonstrating the iterative and embodied nature of creativity in drama pedagogy.

The main contribution of this research lies in the Cognitive Pathway Model of Creativity, which integrates contextual inputs, cognitive processes, and reflective outputs. This model not only extends the theories of Guilford [3], Sawyer [6], and Robinson [7] into a drama-based context but also highlights the role of AI as a reflective partner that fosters metacognitive awareness and creative confidence among pre-service teachers.

Practically, the findings suggest that teacher education programs can benefit from embedding AI-supported reflection checkpoints, encouraging multimodal reflection, and

promoting responsible engagement with digital tools. Such practices can help future teachers develop the creativity, adaptability, and digital literacy required for 21st-century classrooms.

Nevertheless, the study's scope is limited by its small, single-institution sample. Future research should broaden the participant base, explore cross-cultural contexts, and adopt longitudinal designs to examine the sustained influence of AI-assisted reflection on teacher identity and pedagogical innovation.

In conclusion, AI can serve as a constructive and co-creative partner in drama-based teacher education, helping to illuminate hidden dimensions of creative thinking while supporting teacher trainees in becoming reflective, innovative, and ethically informed educators.

REFERENCE

- [1] K. Charmaz, "Constructivist grounded theory," *J. Positive Psychol.*, vol. 12, no. 3, pp. 299–300, 2017.
- [2] A. W. Crapo, L. B. Waisel, W. A. Wallace, and T. R. Willemain, "Visualization and the process of modeling: A cognitive-theoretic view," in *Proc. 6th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 218–226, Aug. 2000.
- [3] J.P. Guilford, J. P. Guilford, "The structure of intellect," *Psychological Bulletin*, vol. 53, no. 4, pp. 267–293, 1956.
- [4] N. Rane, S. Choudhary, and J. Rane, "Education 4.0 and 5.0: Integrating artificial intelligence (AI) for personalized and adaptive learning," *SSRN Electronic Journal*, 2023. [Online]. Available: <https://ssrn.com/abstract=4638365>
- [5] K. Robinson, L. A. Viking, J. Myers, and M. Adams-Budde, *Creative Schools: The Grassroots Revolution That's Transforming Education*. New York, NY, USA: Viking, 2016.
- [6] R. K. Sawyer, "Improvisational cultures: Collaborative emergence and creativity in improvisation," *Mind, Culture, and Activity*, vol. 7, no. 3, pp. 180–185, 2000.
- [7] K. Sawyer, "Extending sociocultural theory to group creativity," *Vocations and Learning*, vol. 5, no. 1, pp. 59–75, 2012.